Quang Truong

truongc4@msu.edu | quang-truong.github.io | quang-truong | quang-cm-truong | Google Scholar

Professional Summary

Doctoral student with 5+ years of experience in software engineering, machine learning, and artificial intelligence. Skilled in designing and deploying scalable ML systems and deep learning solutions. Expertise in **Graph Machine Learning**, **Relational Deep Learning**, **Knowledge Graphs**, and **Large Language Models (LLMs)**, with publications in **NeurIPS**, **AAAI**, and **JMLR**. Strong background in industry research, large-scale model development, and open-source contributions. Experienced across the ML lifecycle, including data processing, model design, training optimization, and distributed computing. Proficient in PyTorch and Python, with solid foundations in algorithms, databases, and high-performance computing (HPC). Seeking a **Research or Engineering Internship (May–Sept 2026)** to advance applied AI research and deliver high-impact, production-ready solutions.

Education

Michigan State University — *East Lansing, MI*
Ph.D. in Computer Science — *Aug. 2024 - May 2028*

Dartmouth College — *Hanover, NH*
M.S. in Engineering Sciences — *Sep. 2022 - Jun. 2024*

Selected Publications

‡ indicates equal contribution.

Peer-reviewed Publications

[1] A Pre-training Framework for Relational Data with Information-theoretic Principles
Quang Truong, Zhikai Chen, Mingxuan Ju, Tong Zhao, Neil Shah, and Jiliang Tang. **2025**
NeurIPS'25: The 39th Annual Conference on Neural Information Processing Systems.

[2] TopoX: A Suite of Python Packages for Machine Learning on Topological Domains
Mustafa Hajij‡, Mathilde Papillon‡, Florian Frantzen‡, ... **Quang Truong**, and ... Nina Miolane. **2024a**
The Journal of Machine Learning Research (Volume 25).

[3] Weisfeiler and Lehman Go Paths: Learning Topological Features via Path Complexes
Quang Truong and Peter Chin. **2024b**
AAAI'24: The 38th Annual AAAI Conference on Artificial Intelligence.

[4] Not All Data Matters: An Efficient Approach to Multi-Domain Learning in Vehicle Re-Identification
Quang Truong and Bo Mei. **2021**
ITSC'21: The 24th IEEE International Conference on Intelligent Transportation Systems.

Open-source Projects

TopoX
- Provides reliable and user-friendly building blocks for computing and machine learning on topological domains that extend graphs.
- Adopted open-source software engineering best practices with comprehensive testing, documentation, and CI/CD pipelines.
- Starred 500+ times across 3 different libraries on GitHub.
- Published at the Journal of Machine Learning Research (JMLR).

Skills

Programming	Python (Expert), C/C++ (Advanced), Java (Proficient), R (Proficient), MATLAB (Proficient)
Machine Learning	Graph Machine Learning, Relational Deep Learning, Knowledge Graphs, Recommender System, Large Language Models, Computer Vision, Data Mining
ML Frameworks	Pytorch, Pytorch Geometric, Pytorch Frame, NetworkX, TopoX, Pytorch Lightning, LangChain, Hugging Face, Scikit-learn, Matplotlib, Numpy, Pandas
DevOps & Infrastructure	WandB, Docker, Singularity, Pytest, Amazon Web Services (AWS), Google Cloud Platform (GCP), Kubernetes, CI/CD, Monitoring, Distributed Systems
Software Engineering	Spring, Agile, JIRA, SCRUM, Git, Linux/Unix, MongoDB, DuckDB, MySQL, Unit Testing, Performance Optimization, Automation, Algorithm Design, Data Structures, System Design

Research Experience

DSE Lab, Michigan State University *East Lansing, MI*
*Doctoral Research Assistant (Advisor: **Jiliang Tang**)* *Aug. 2024 - Present*

- Develop scalable transformers and characterize necessary conditions for the models to be strong Link Predictors (**Snap Research Collaboration**).
- Design and implement distributed training pipelines for large-scale graphs, optimizing performance across multiple GPUs and ensuring scalability for enterprise-scale data.
- Create automated testing frameworks and continuous integration pipelines to ensure code quality and system reliability across research projects.
- Develop specialized pre-training procedures for graph machine learning on relational databases, thereby enhancing predictive modeling for node-level classification, link prediction, and temporal forecasting tasks (**Snap Research Collaboration**).

LISP Lab, Dartmouth College *Hanover, NH*
*Graduate Research Assistant (Advisor: **Peter Chin**)* *Sep. 2022 - Jun. 2024*

- Introduced a novel approach to Topological Deep Learning (TDL) leveraging simple paths, significantly enhancing graph expressivity and surpassing the theoretical limitations of the 3-WL test.
- Developed Path Isomorphism Network, a state-of-the-art TDL model, which delivered outstanding performance on benchmark datasets: PROTEINS (78.8%), NCI1 (85.1%), NCI109 (84.0%), and IMDB-B (76.6%).
- Developed an anomaly detection pipeline for human behavior analysis using geospatially-tagged data, as part of the **IARPA-HAYSTAC** funded project.

Vision Lab, University of Illinois at Chicago *Remote*
*Research Intern (Advisor: **Wei Tang**)* *May 2021 - Aug. 2021*

- Designed a context-aware 3D object detection architecture that captures both object-wise and object-context relationships, improving spatial understanding.
- Boosted model performance through cross-modality alignment, improving the feature representation of objects.

ML Lab, Texas Christian University *Fort Worth, TX*
*Undergraduate Research Assistant (Advisor: **Bo Mei**)* *May 2019 - May 2021*

- Developed an advanced Vehicle Re-identification pipeline utilizing Generative Adversarial Networks (GANs) for adaptive domain learning, resulting in a 10% improvement in mean Average Precision (mAP) over the baseline.
- Engineered a novel filtering algorithm that reduces the training data requirement by 25% while maintaining model performance, significantly improving computational efficiency.

Presentations

Invited Presentations

- Topologically-driven Neural Networks, *HPCC Special Seminar: Decoding Computational Challenges, the Advanced Institute of Interdisciplinary Science and Technology - HCMUT, VNU-HCM*. **2024**

Conference Presentations

- Weisfeiler and Lehman Go Paths: Learning Topological Features via Path Complexes, *AAAI'24: The 38th Annual AAAI Conference on Artificial Intelligence*. **2024**
- Not All Data Matters: An Efficient Approach to Multi-Domain Learning in Vehicle Re-identification, *ITSC'21: The 24th IEEE International Conference on Intelligent Transportation Systems*. **2021**

Professional Services

- Conference Reviewer, *Learning on Graphs Conference (LoG)*. **Fall 2025**
- Program Committee Member/Conference Reviewer, *Annual AAAI Conference on Artificial Intelligence (AAAI)*. **Fall 2025**
- Conference Reviewer, *Conference on Neural Information Processing Systems (NeurIPS)*. **Summer 2025**
- Journal Reviewer, *IEEE Transactions on Big Data (TBD)*. **Summer 2025**
- Conference Reviewer, *International Conference on Learning Representations (ICLR)*. **Fall 2024**
- Journal Reviewer, *ACM Transactions on Knowledge Discovery from Data (TKDD)*. **Fall 2024**

Honors

- NeurIPS 2025 Scholar Award, *Neural Information Processing Foundation*. **2025**
- VEF Fellow, *VEF 2.0 Program, VEF Fellows and Scholars Association*. **2021**
- Vingroup Scholarship Recipient, *Master's & Ph.D. Scholarship Program, VinUniversity*. **2021**
- Bronze Medal, *ACM-ICPC South Central USA Regional Contest [URL]*. **2019**